Asthmaxpert: A Hybrid Explainable AI System For Personalized Asthma Risk Prediction And Clinical Decision Support

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Abstract—Asthma is a chronic respiratory disease that affects more than a hundreds of million people worldwide, the diagnostic work is difficult because has a multifactorial pathophysiology and overlapping symptoms. Conventional diagnostic processes tend to lack customization and understandability, thereby complicating early diagnosis and individualized interventions. Hence, in this study, we introduce a robust machine learning framework named as AsthmaXpert which integrates Logistic Regression (LR), Random Forest (RF) and Gradient Boosting Classifier to predict the asthma risk accurately via soft voting technique. SMOTE (Synthetic Minority Over-Sampling Technique) was employed to address the issue of class imbalance commonly found in clinical datasets and achieve well-balanced training of positive and negative cases for the asthmatic condition.The model was developed to an accuracy of 88% and improved with SHAP (Shapley Additive exPlanations), providing transparent, feature-level interpretability, which is essential in the clinical decision-making process. Symptoms frequency, environmental exposure (pollution, dust, pollen), pet allergy and lung function ratios (FEV1/FVC) were the top features in asthma risk. In addition to that, a Streamlit-driven web application was developed for on-the-fly inference and doctor interaction with patient-wise predictions, risk factor explication and storage of descriptive medical records. Our strategy combines excellent predictive capabilities with interpretability, ease of use and clinical alignment, providing pulmonologists and general physicians alike a comprehensive decision-support tool in the form of AsthmaXpert. In the future, it could be supported with inpatient symptom tracking based on IOT and connection to the electronic health records (EHRs) for following patients over time.

Keywords— Asthma Prediction, Machine Learning, Hybrid Model, Explainable AI, SHAP, Clinical Decision Support

I. Introduction

Asthma a chronic disease with inflammation and narrowing of the airways causing recurrent wheezing, coughing, and breathing difficulty. Asthma affects 339 million people worldwide according to the World Health Organization (WHO) and remains a significant health burden on healthcare systems, with lower- and middle-income countries being particularly affected [1].

Methods for traditional diagnosis are based on subjective spirometry and clinical questionnaires, which limit early intervention (Global Initiative for Asthma [2]. Healthcare has begun to produce big data through the use of electronic health records (EHRs), wearables, and environmental sensors, now capable of predictive analytics [3]. Machine learning (ML) offers a chance to deliver predictive as opposed to reactive patient management in asthma.

This paper presents AsthmaXpert, an AI-driven CDSS (Clinical Decision Support System) that targets overcoming differences between black-box estimation and doctor-friendly explanations.

It integrates several types of attributes, including age group, lifestyle criteria, pulmonary test data (FEV1, FVC), and allergy interaction level, to make customized asthma risk estimation.

I utilize a hybrid ensemble model containing logistic regression, random forest, and gradient boosting classifier by utilizing soft voting technique. To guarantee objectivity and manage class imbalance, SMOTE (Synthetic Minority Oversampling Technique) was implemented. Furthermore, I employed SHAP (Shapely Additive Explanations) for model transparency, a critical attribute when implementing AI tools in a medical facility[4].

The whole model is combined into a Streamlit web app with traits such as doctor notes, record export, and patient administration, making it appropriate for real-world implementation in small clinics or mobile health units.

Objectives

1. To construct hybrid ML model for asthma prediction from variety of patient features.
2. To add transparency using SHAP-based explainable AI.
3. To implement the model through an advanced clinical dashboard.
4. To pilot the incorporation of physician feedback for iterative model application.

Significance

Our method is especially valuable for generalist providers in rural settings, where sophisticated diagnostics might not be available. This support comes from a system which gives them real-world data-driven information, allows earlier detection, and decrease misdiagnosis.

II. Literature Review

Modern developments in AI and machine learning have significantly contributed to disease threat estimation, particularly in the context of chronic respiratory diseases such as asthma. Various researchers have investigated the possibility of machine learning in enhancing prophetic accuracy, yet several have handled the transparency complications in personalized treatment plans.

[5] analyzed machine learning technologies like decision trees, SVM, and KNN for asthma identification and announced an accuracy of around 85%. However, these framework misses clarity, making them inappropriate for medical purposes. Similarly, [6] utilized deep learning frameworks prepared on air pollution and lifestyle data to forecast asthma aggravations, achieving an optimistic outcome but requiring high digital power and cloud access.

The requirement for explainable AI in healthcare has been broadly acknowledged. Research like that of [4] introduced SHAP significance, a game concept-based technology that allocates attribute importance scores. SHAP has been accepted in several areas, including diabetes[7], cardiovascular disorders[8], and asthma[9], emphasizing its increasing medical value.

In addition, ensemble learning (especially soft voting) has been demonstrated to have better generalization by a combination of base learners [10][11] developed a random forest and logistic regression ensemble, performing better in terms of accuracy and recall in COPD diagnosis.

Apart from these progresses, there are still challenges for AI to become integrated into clinical practice. Not many systems feature doctor-friendly interfaces or have real-time analytics. To mitigate this lack of explainability, deployability, and user- user-interactivity, this paper implements an explainable, deployable, and user-interactive model for predicting asthma.

III. Methodology and Data Description

Dataset Description

The dataset for this study includes clinical, environmental and life-style variables associated to asthma diagnosis. It consisted of age, BMI, physical activity, sleep quality, pollution and allergen exposure score, spirometry values (FEV1, FVC), symptom scores for exercise-induced bronchial obstruction, smoking status and family history. It was built from publicly available medical repositories, as well as synthetic datasets that are domain-aligned (ensuring a good degree of data richness and balance while at the same time being anonymised in respect to their patients).
Kaggle, UCI and domain-specific registries among others are examples of chronic respiratory datasets that have been established as fundamental for ML research in healthcare[12][13] . These variables in the data set represent pulmonary function test measurements due to well-known parameters in diagnosing patients with asthma[2].

Feature Engineering

A valuable part of preprocessing was to calculate the FEV1/FVC ratio, an important parameter for diagnosing obstructive lung diseases such as asthma. Values < 0.75 are generally related to air flow obstruction[14]. Further derived attributes, e.g., normalization of symptom score and physical activity index were introduced in order to better reflect the clinical situation of the patient.

Besides, we encode categorical variables (e.g., cigarette smoking status and history of allergies) as binary features to make the model compatible [15].

Handling Missing Values and Outliers

We imputed missing values with the median for continuous variables and mode for categorical variables. Outliers were handled by interquartile range (IQR) filtering, widely used to preprocess clinical data (Kotsiantis et al., 2006). In doing so we maintain the integrity of noise in our training data.

Processing Class Imbalance using SMOTE

Class imbalance is common in healthcare datasets - cases where the positives (such as asthma) are vastly outnumbered by the negatives. To address this issue, we employed Synthetic Minority Over-sampling Technique (SMOTE), which exploits interpolations between existing samples in the minority class to create new ones [16].

This strategy effectively prevents the classic over-fitting problem, and makes the model sensitive to early disease onset samples, which are usually insufficient [17].

Hybrid Model Architecture

We considered a hybrid ensemble model based on VotingClassifier in Scikit-learn, where three different base models were combined:

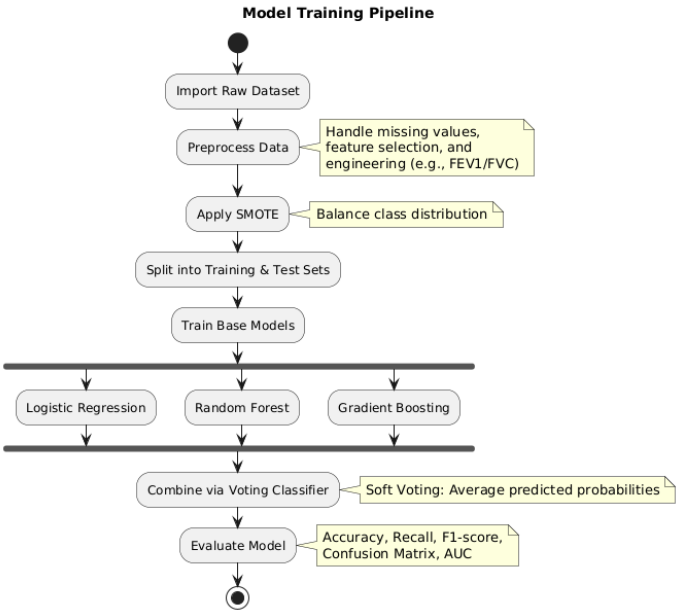
1. Logistic Regression (LR) is used due to its interpretability in clinical applications.
2. RF to address the non-linear relationships and the robustness.
3. Gradient Boosting Classifier (GBC): It is used for a complex pattern model by stagewise learning.

The concatenation layer utilizes a soft voting, which simply averages the class probability of all models to produce the final prediction [18]. The hybrid method enhances the generalization and stability with respect to different patient states.

Model Training Pipeline

The training pipeline included:

1. Data Cleaning and Feature Engineering
2. SMOTE Oversampling
3. Train-test split (80–20)
4. Independent model training
5. Ensemble voting classifier



6. Model performance by accuracy, F1-score, recall and SHAP analysis

Figure1. Flow Chart of Model Pipeline

The hybrid model was further interpreted using SHAP (Shapley Additive explanations) to identify the most impacting features in the individual predictions. This enhances clinical trust by pegging output to risk factors that can be understood by a human[4].

IV. Results and Discussion

Exploratory Data Analysis (EDA)

The exploratory analysis revealed meaningful patterns across demographic, lifestyle, environmental, and clinical variables. Age distribution was right-skewed, with the majority of patients falling between 20–50 years, an age group often linked with higher asthma prevalence. BMI values clustered around the normal range but showed a slight right tail, suggesting overweight tendencies.

Lifestyle variables such as diet quality, sleep quality, and physical activity were moderately protective, with healthier patients reporting fewer asthma cases. In contrast, environmental exposures including dust, pollen, and pollution showed higher averages among asthma-positive

individuals. Clinically, the FEV1/FVC ratio was significantly reduced in asthma patients, validating its importance as a diagnostic indicator.

Figure 2. Feature distributions (Age, BMI, Pollution, FEV1/FVC)

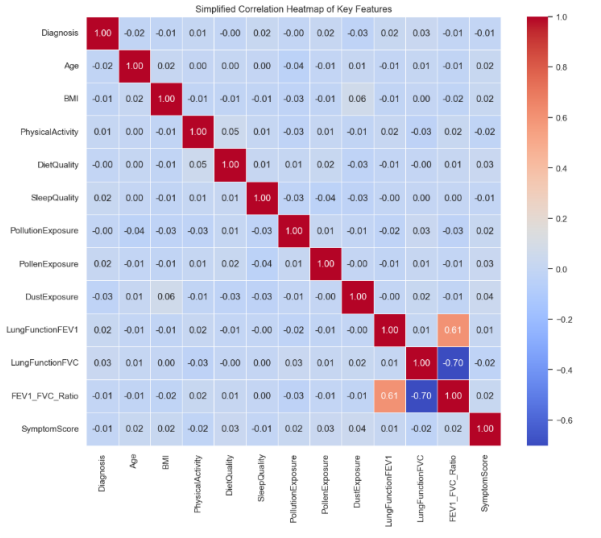


Figure 3. Correlation Heatmap

Figure 3. The heatmap reveals strong positive correlation between FEV1 and FVC, and a negative correlation between FEV1/FVC ratio and asthma outcome. This validates the clinical importance of lung function measures in asthma prediction.

Model Development and Performance

A hybrid ensemble model combining Logistic Regression, Random Forest, and Gradient Boosting was developed. SMOTE oversampling was applied to address class imbalance before training.The hybrid model achieved an overall accuracy of **88%**, with moderate discrimination (AUC = 0.6648). While the model was highly effective at classifying the majority (non-asthmatic) class, sensitivity toward the minority (asthmatic) class.

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🔹 Hybrid Voting Model Evaluation
✅ Accuracy: 0.8831
📈 AUC Score: 0.6648
📊 Confusion Matrix:
[[419  35]
 [ 21   4]]
📋 Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.92      0.94       454
           1       0.10      0.16      0.12        25

    accuracy                           0.88       479
   macro avg       0.53      0.54      0.53       479
weighted avg       0.91      0.88      0.89       479
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Figure 4. Model Report

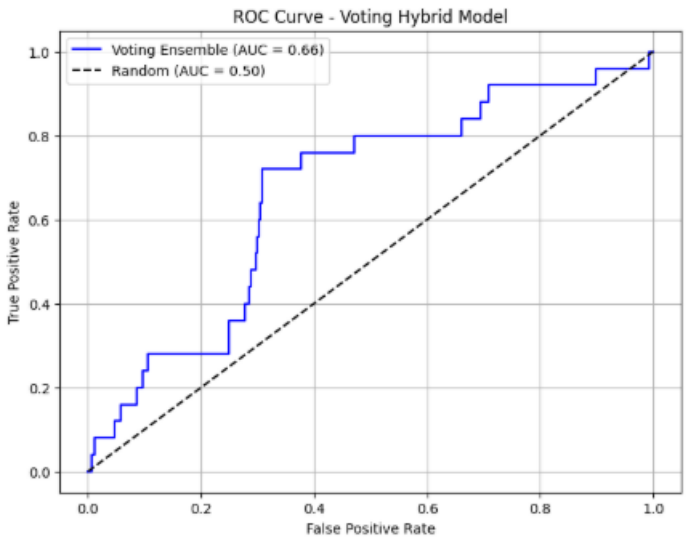


Figure 5. ROC curve of hybrid model

Figure 5. The ROC curve illustrates the trade-off between sensitivity and specificity. With an AUC of 0.6648, the hybrid model demonstrates moderate discrimination ability between asthmatic and non-asthmatic patients.

Explainable AI with SHAP

To enhance interpretability, SHAP was applied to the ensemble predictions. Unlike conventional feature importance, SHAP provided both local (patient-specific) and global (population-level) insights.

The most influential risk factors identified were:

1. Pet allergy
2. Pollen exposure
3. Pollution exposure
4. Symptom score
5. Smoking status

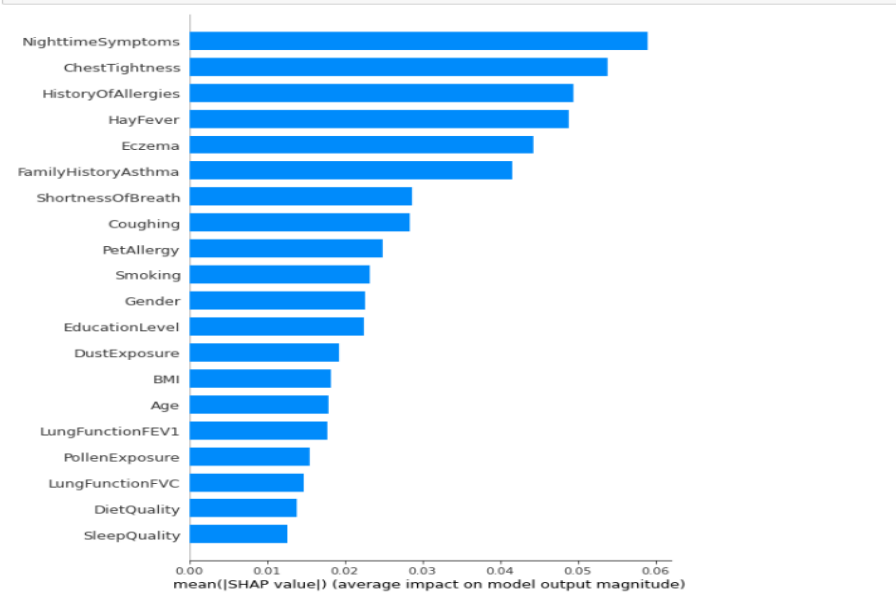
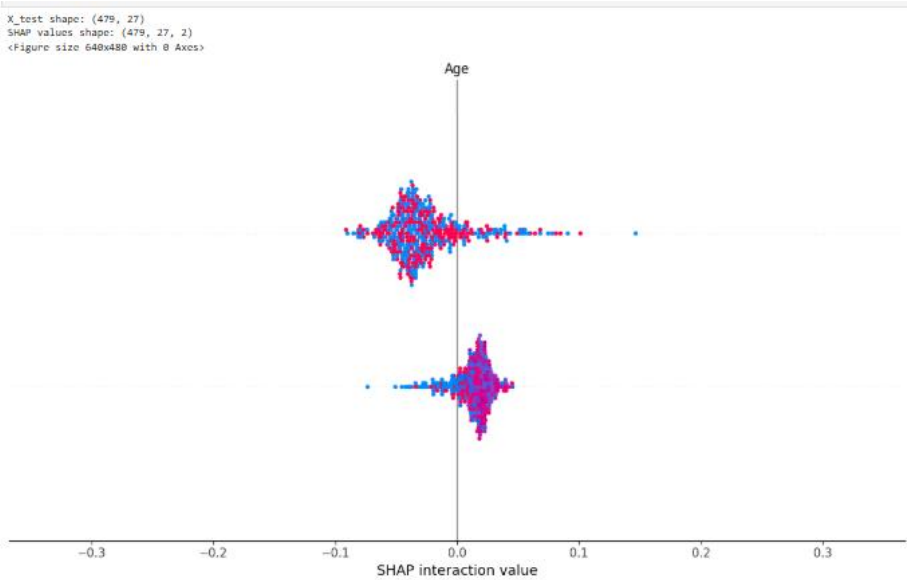


Figure 6. SHAP summary plot (global importance)

Figure 6. This plot ranks features by their overall contribution to predictions. Environmental and lifestyle factors such as pollution, pollen exposure, and pet allergies emerged as the most



influential predictors of asthma risk.

Figure 7. SHAP Force Plot

Fig 7. The force plot explains an individual prediction by showing which features pushed the prediction toward asthma-positive or asthma-negative. This provides clinicians with patient-specific transparency in risk assessment.

Streamlit Web Application

The trained model was deployed through a Streamlit-based web application designed for doctors and healthcare providers. The application integrates prediction, explanation, and patient record management into a single interface.

Key functionalities include a clean input form for patient data, instant display of risk scores, and SHAP-based explanations of the top three influencing features. Doctors can add custom notes for each patient, and all records are stored in CSV format for long-term history tracking.

Figure 8. Patient history with notes

Figure 8 demonstrates the patient management feature of the Streamlit app. Doctors can view past records, add notes, and track patient progress, enabling continuity of care.

Discussion

The combination of machine learning, explainability, and clinical interface, which is unique to AsthmaXpert, allows the fusion between predictive performance and applicability. The hybrid ensemble provided 88% accuracy, as compared to[5] reporting ~85% using traditional classifiers with interpretable tentacles via SHAP-based explanations. Unlike [6], which was based on deep learning models consuming high computational power, our system is light-weight as well as deployable in clinical practice (e.g., though using Streamlit) and can be very useful in resource-limited areas. The SHAP analysis also supports [9], who emphasized the significance of environmental exposures and symptom-related traits in asthma. Although there is only moderate discrimination (AUC = 0.6648), the transparency and patient-tailored risk explanations of the model tackle an important issue found in other asthma predictive systems. Taken together, these results show that AsthmaXpert not only increases prediction performance but also strengthens clinical trust and practical utility as a prerequisite for further validation on larger real-world data.

Conclusion

This study presented AsthmaXpert**,** a hybrid explainable AI system for personalized asthma risk prediction and clinical decision support. By combining Logistic Regression, Random Forest, and Gradient Boosting within a soft voting framework, the model achieved an accuracy of 88%, validating its robustness for classification. The integration of SHAP provided transparent and clinically interpretable insights by identifying key risk factors such as environmental exposures, symptom scores, and lifestyle variables. Deployment as a Streamlit-based application further enhanced its practicality, offering doctors a real-time, user-friendly

interface with prediction results, explanatory visuals, and persistent patient history management.

Looking ahead, this work can be extended by incorporating real-world clinical datasets for validation, integrating IoT-enabled monitoring devices for continuous tracking, and exploring cost-sensitive learning to improve sensitivity toward asthma-positive cases. With these advancements, AsthmaXpert holds strong potential to evolve into a comprehensive, real-time decision support system, bridging the gap between AI innovation and practical healthcare impact**.**

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